

Group: _____ **Section:** _____

Member 1: _____ **BUID:** _____

Member 2: _____ **BUID:** _____

Member 3: _____ **BUID:** _____

Member 4: _____ **BUID:** _____

Project 2 Launch: Bootstrapping and Study Selection

Purpose

Today your group will launch Project 2 and work directly on the **Deliverable 1 outline**. By the end of lab, your shared Google Doc should contain a complete first draft covering both the simulation study and real-data analysis.

Shared Google Doc Setup

1. Create one Google Doc for the group and name it *P2_{GroupNumber}outline*.
1. Share the document with all four group members and the instructor/TA with editing access.
2. Add all four names and BUIDs at the top. Use headings that match the Deliverable 1 outline.
3. Work in the same document throughout lab; do not create four separate drafts.

Group-of-Four Roles

- **Member 1 – Project lead:** background, motivation, and final integration.
- **Member 2 – Simulation lead:** selected study 1, violation, data-generating process, and simulation plan.
- **Member 3 – Simulation lead:** selected study 2, violation, data-generating process, and simulation plan.
- **Member 4 – Real-data and editor lead:** dataset, variables, limitations, timeline, and final quality check.

In-Class Work

- (A) **Project orientation.** Review the Project 2 overview and rubric together; identify the required simulation and real-data components.
- (B) **Bootstrap practice.** Implement or review a bootstrap confidence interval for a regression quantity and record what the code returns.
- (C) **Study selection.** Choose two studies (A–D), explain the assumption violations, and justify why they are useful to compare.

- (D) **Real-data plan.** Select a dataset, response Y , primary predictor X , source, and expected assumption violations.
- (E) **Integrate the outline.** Add sample sizes, repetitions, data-generating processes, planned outputs, timeline, and roles to the shared document.
- (F) **Check-in.** Ask the instructor/TA to review the two selected studies and the connection between the simulations and real-data analysis.

Deliverable

Submit the completed Project 2 outline through Gradescope or Blackboard by Sunday, October 18 at 10:00 PM. Your shared Google Doc must contain:

- project title, motivation, and all four group members;
- two selected simulation studies and a plan for each;
- the two core research questions and any group-specific hypotheses;
- the real-data source, variables, limitations, and expected violations; and
- timeline, responsibilities, and planned simulation/analysis outputs.

Before submitting, confirm that every member has contributed, the document is readable without comments, and the final link is accessible to the instructor/TA.